Architectural Particular Specification

CAST DECKS AND UNDERLAYMENT

SECTION 2.1

Rev	Date	Description
0	14 APR 2022	First Issue

Main Contract for
Proposed Data Centre Development
at 2-10 Tai Yuen Street,Kwai Chung, N.T.
Andrew Lee King Fun & Associates Architects Limited

SECTION 2.1

CAST DECKS AND UNDERLAYMENT

PART 1 GENERAL

1.01 SUMMARY OF WORK

A. Section Includes: The Work of this Section includes, but is not limited to, the furnishing and installation of the following :

1. Surface preparation and repair of substrate.
2. Provide and install reinforced waterproof, cement/sand and roof screeding over existing concrete substrate.
3. Temporary fire protection during installation of flammable SBD products and torch-on APP modified bitumen roof membrane.

1.02 REFERENCES

A. British Standards (BS):

1. BS 3797:1990 - Specification for Light Weight Aggregates for Masonry Units and Structural Concrete.
2. BS4449: 1988 - Specification for Carbon Steel Bars for the Reinforcement of Concrete.
3. BS6319: All applicable parts : Test of Resin and Polymer/Cement Compositions for Use in Construction (date varies).
4. BS 8000: Part 9: 1989 - Code of Practice for Cement/Sand Floor Screeds and Concrete Floor Toppings
5. CP 204 : Part 2 : 1970 - In-situ Floor Finishes

B. American Concrete Institute (ACI) :

1. ACI 301: Specifications for Structural Concrete for Buildings.
2. ACI 305R: Hot Weather Concreting.
3. ACI 306R: Cold Weather Concreting.
4. ACI 308: Standard Practice for Curing Concrete.
5. ACI 315 : Details and Detailing of Concrete Reinforcement.

C. American Society for Testing and Materials (ASTM):

1. ASTM A185 : Welded Steel Wired Fabric for Concrete Reinforcement.
2. ASTM A615: Deformed and Plain Billet - Steel Bars for Concrete Reinforcement.
3. ASTM C31 - 91: Making and Curing Concrete Test Specimens in the Field.
4. ASTM C33 - 93: Concrete Aggregates.
5. ASTM 39 - 93a: Compressive Strength of Cylindrical Concrete Specimens.
6. ASTM C94 - 94: Specification for Ready-Mixed Concrete Specimens.
7. ASTM C143 - 90a: Test Method for Slump of Portland Cement Concrete.
8. ASTM C150 - 94: Specification for Portland Cement.
9. ASTM C172 - 90: Method of Sampling Freshly Mixed Concrete.

10. ASTM C881 - Epoxy Resin Base Bonding Systems for Concrete.

D. Local Codes:

1. Design Manual - Barrier Free Access 1997

1.03 QUALITY ASSURANCE

(NOT USED)

1.04 PERFORMANCE REQUIREMENT

A. Floor Screeds

1. Reinforced waterproof cement/sand screeding shall have a minimum compressive strength requirement: 52N/mm² after 28 days.

2. No cracking or spalling of lightweight screed permitted under design loading conditions.

B. Non-Structural Concrete

1. Refer to Consulting Engineers specification for material standards.

2. No cracking or spalling of concrete permitted under design loading conditions.

1.05 ENVIRONMENTAL REQUIREMENTS

(NOT USED)

1.06 ENVIRONMENTAL, CLIMATIC & PSYCHOMETRIC CONDITIONS

A. Hot Weather Concreting: Perform work in accordance with ACI 305R.

1. When hot weather conditions exist that would seriously impair the quality and strength of concrete, place concrete in compliance with ACI 305R and as herein specified.

2. Temperature of concrete at time of placing: Not to exceed 32.5°C. Maintain an accurate reading thermometer at the job site to check temperature of concrete. Reject concrete before placing if temperature of concrete exceeds 32.5°C.

3. Execute special precautions to protect fresh concrete before and during finishing when the rate of evaporation of surface moisture from concrete exceeds 1kg/m² per hour. Determine rate of evaporation in accordance with ACI 305R. Provide special precautions as required:

a. Cool ingredients before mixing to reduce concrete temperature at time of placement. Mixing clean, potable water may be chilled, or chopped ice may be used to control the concrete temperature provided the water equivalent of the ice is calculated to the total amount of mixing water.

b. Dampen subgrade and forms.

c. Cover reinforcing steel with water-soaked burlap so the steel temperature will not exceed the ambient air temperature immediately before embedment in concrete.

1.07 SUBMITTALS

A. Propose sub-contractor specializing in installation of cementitious systems with ten (10) years minimum experience on projects of a similar size and complexity.

B. Where reference is made to products from various manufacturers, submit a list of the proposed products and confirm compatibility between all elements/materials as appropriate.

C. Alternatively product from a single source supplier is permitted. Confirm compatibility between all elements/materials as appropriate.

1.08 PRE-INSTALLATION CONFERENCE

(NOT USED)

1.09 MOCK-UPS

This part shall be read in conjunction with Clause 8.03 of Specification Preliminaries for the requirements for prototypes and trial areas.

1.10 TESTING

A. Before commencing any Screeding Works, the Contractor must demonstrate on Site that all adhesive and materials are fully compatible with each other, including as-installed waterproofing system. A mock-up panel of size not less than 2m x 2m on selected screeding works (to be selected by Architect) shall be carried out by the contractor for the inspection and approval by the Architect prior to bulk ordering of the selected screeding. No work shall be undertaken when the surface moisture exceeds the permissible, as tested by the moisture testing equipment on Site.

B. Product Test Reports: Based on evaluation of comprehensive tests performed by [manufacturer and witnessed by a qualified testing agency], for concrete floor topping.

C. In-Situ Pull Out Test

D. In – Situ Hammer Tapping Test

E. Tolerance Test

1. Test the level and surface irregularity of the Screeding System to verify that they are within the tolerances stated in this section using the method described in Appendix A of BS 8204 Part 2. The Contractor shall conduct this test at the frequency of one test per $100m^2$ of Screed.

F. Testing and inspecting of completed applications of concrete floor toppings shall take place in successive stages, in areas of extent and using methods as follows:

1. Sample Sets: At point of placement, a set of 3 molded-cube samples shall be taken from the topping mix for the first $100m^2$, plus 1 set of samples for each subsequent $500m^2$ of topping, or fraction thereof, but not less than 6 samples for each day's placement. Samples shall be tested according to ASTM C 109/C 109M for compliance with compressive-strength requirements.

2. Concrete floor topping shall be tested for delamination by dragging a steel chain over the surface.

3. Concrete floor topping shall be tested for compliance with surface flatness and levelness tolerances.

4. Remove and replace applications of concrete floor topping where test results indicate that it does not comply with specified requirements.

1.11 WARRANTY

A. This part shall be read in conjunction with Clause 8.04 of Specification Preliminaries for the list of material/ system requiring joint written guarantees.

B. Provide Ten (10) years warranty for all screeding work against material defects, workmanship, cracking and crazing.

C. The Management Contractor and the Applicator shall warrant that all products, materials and items are suitable, and can be maintained in the various climatic and psychometric conditions encountered without cracking, delamination, or material defects.

D. Nothing shall be done by the Management Contractor which will void any Manufacturer's / Supplier's Warranty.

1.12 SPARE PARTS

This part shall be read in conjunction with Clause 8.05 of Specification Preliminaries for the list of spares and spare parts.

PART 2 PRODUCTS

2.01 APPROVED MANUFACTURERS/SUPPLIERS

A. Subject to compliance with the Project Contract Document Requirements, provide cementitious screed and ancillary materials from the following or equivalent approved by the Architect.

1. TECH LINK CONSTRUCTION ENGINEERING LIMITED

   Unit 802, 8/F, Laford Centre,

   No. 838 Lai Chi Kok Road,

   Cheung Sha Wah, Kowloon,

   Tel: (852) 3701 1817 / 9316 8906 Fax: (852) 3701 1818

   Email: kenuskwong@techlinkcon.com.hk; or

2. Equivalent approved by Architect

2.02 MATERIALS

To provide the following cementitious screed and ancillary materials or equivalent approved by the Architect:

A. **Sand Cement Screed**

Cement/Sand Screed

1. Screed Materials:

   a. Screeds generally to consist of cement and sand 1:3. Use the minimum clean potable water content required for workability.

   b. Screeds over 40mm thick shall be mix to 1 part cement, 1.5 parts granite fines and 3 parts course aggregate graded 10 mm down with at least 75% being retained on a 5 mm B.S. sieve.

   c. Floor screeds are generally to be laid in accordance with CP 204:pt 2 and B.S.8000: part 9.

2. Cement : Portland Cement ordinary or rapid-hardening with specific surface of 225 m^2/kg and conforming to BS12.

3. Admix and Bonding Aid: Polymer, etc. admixture by Proprietary Manufacturer.

   - "PCI Latex" (formerly known as "PCI Emulsion") modified synthetic resin dispersion base bonding slurry coat/spatterdash coats and bedding mortar or approved equivalent for Flat Roof and Podium Deck, All internal wet area e.g. Lavatory / Plant room / Pump room / Refuse room and external wall with rendering finishes, etc.:

<u>The performance requirements of "PCI Latex" (formerly known as "PCI Emulsion") modified synthetic resin dispersion base bonding slurry coat/spatterdash coats and bedding mortar:</u>

a. Chemical nature - Aqueous dispersion of a polymer based on: acrylic polymers;

b. Vapour pressure: 23 hPa;

c. Information on water vapour pressure: 23 mbar (20°C);

d. pH value: 8.3 – 9.2 (20°C);

e. Possible Hazards (according to Directive 67/548/EWG or 1999/45/EC): No specific dangers known, if the regulations / notes for storage and handling are considered;

f. Can be used to modify mortar and render;

g. Plasticising, better workability and easier application of mortars and renders;

h. Help increase the wear resistance of mortar, less abrasion, thus longer solutions;

i. Improve bending tensile strength; help achieve a stress-free setting process, even with large areas;

j. Crack-free hardening, reduces movement in the mortar;

k. Minimum Flexural strength of BASF PCI EMULSION modified render according to EN 196 Part 1 : 7.3 N/mm²;

l. Minimum Compressive strength of BASF PCI EMULSION modified render according to EN 196 Part 1 : 23 N/mm² but less than 30 Mpa;

m. Capillary water absorption coefficient of BASF PCI EMULSION modified render according to EN 13057: 0.70 kg / (m².h0.5);

- "BASF Febond SBR" modified cement bonding slurry coat/spatterdash coats and bedding mortar or approved equivalent for Water Tank:

The performance requirements of "BASF Febond SBR" modified cement bonding slurry coat/spatterdash coats and bedding mortar:

				Normal Dose Range			
Quantity of latex in litres per 50kg cement used				10	12.5	15	20
Adhesion to concrete:	1W27D:	DRY	N/mm²	2.0		2.1	3.4
Test Method 1	1W20D7W:	WET	N/mm²	1.0		0.9	1.4
	1W20D63W:	WET	N/mm²	1.4		1.8	2.1
Adhesion to steel:		DRY	N/mm²	2.0			1.6
Test Method 1		WET	N/mm²				1.3
Slant Shear Adhesion (BS 6319 : Part 4) (Mortar / Febond SBR mortar using Febond SBR bonding slurry).							
Test Method 2:		N/mm²		31			
Tensile strength (BS12):		DRY	N/mm²				4.3
		WET	N/mm²				3.9
Compressive strength (BS 1881:Part 4) :		DRY	N/mm²	0			
Flexural strength (BS 4551):		DRY	N/mm²		6.2		10.6
		WET	N/mm²				9.6
Effect of chemicals on Flexural Strength after 6 month immersion:		DRY					
Untreated		N/mm²			6.2		
10% Potassium Hydroxide		N/mm²			12.3		
10% Magnesium Sulphate		N/mm²			6.3		
5% Lactic Acid		N/mm²			8.0		
10% Sucrose		N/mm²			9.2		
5% Hydrochloric Acid		N/mm²			2.2		
20% Ammonium Nitrate		N/mm²			4.8		
Petroleum Spirit		N/mm²			7.5		

Effect of Extremes of Temperature:				
Flexural strength:				
Untreated	N/mm²			10.6
After 60 freeze-thaw cycles in 10% brine at -18°C	N/mm²			10.4
After 1 year at 70°C	N/mm²			14.3
Adhesion to Concrete:				
Untreated	N/mm²			3.4
After 6 months at 70°C	N/mm²			2.6
Thermal Expansion (Linear):				
from -20°C to +20°C	(Coeff x 106)		12.8	
from +20°C to +60°C	(Coeff x 106)		12.9	
Shrinkage during drying:				
Water / cement ratio		0.34	0.34	0.30
% shrinkage		0.02	0.01	0.01
Water Vapour Permeability (BS 4016 / BS 3177 11mm thick):	g / m² / 24 hrs	38.1	3.9	1.9
Water Penetration (30 metres per head of water was applied to one face of a mortar 15mm thick: (i) with Febond SBR in mortar (ii) using Febond SBR / cement sealing coats on unmodified mortars	kg / m² / 24 hrs	35	0	0

Other Performance:

Description	Values
pH	10.5
Specific gravity	1.01
Freeze-thaw stability	Withstands at least 5 freeze thaw cycles, but insides storage above 0°C recommended.
Particle size	0.17 micron*
Stabilization	Non-ionic
Butadiene content	45% by weight of the polymer
Antioxidant	Yes
Bacteriocide	Yes
Antifoam	Yes
Minimum film forming temperature	About 1°C
Shelf life (interior storage from above 0°C to 25°C	About 2 years, but agitate before use after prolonged storage
(* A cubic mm of dried film is made up from more than 125,000,000,000 particles	

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B. **Concrete for Non-Structural Work**

1. Non-Structural Concrete: Refer to consulting Engineers specification for requirements of materials used in concrete work.

C. **Floor Hardener: Proprietary Heavy Duty & Dust-Free Floor Hardener "KORODUR 0/4 dry shake hard aggregate" and "Korodur Korotex" Penetrating Sealer / Water Repellent for pump/ plant room:**

1. KORODUR 0/4 Mixing ratio:

Type of vehicular traffic	KORODUR 0/4 aggregate kgs / sqm	Cement kgs / sqm	Total mix KORODUR 0/4 + cement kgs / sqm
Light traffic (UP to 10 ton)	2	1	3
Medium truck traffic (Up to 20 ton)	3	1.5	4.5
a. Frequent forklift traffic b. Heavier truck traffic (Up to 40 ton)	5	2.5	7.5
The most heaviest traffic	7	3	10

2.

Description	Technical Data of KORODUR 0/4 or VS 0/5 Hard Aggregate
Flexural strength N/mm^2 (average value)	11.7 (DIN 1100)
Compressive strength N/mm^2 (average value)	91.4 (DIN 1100)
Non-premixed hardener	Yes
Aggregate size and treatment	0-4mm in diameter, have been undergone electro-metallurgical smelting processes
Wear cm^3 / 50cm^2 (average value)	4.50 (DIN 1100)
Bulk density kg / dm^3	1.68 (DIN 1100)
Granulometric composition	acc. to DIN 4226: Part 3
Shelf Life	Unlimited
Quality and Factory Control	acc. to EN 13813
Slip Property	'R12' and 'R13' according to DIN 51130 slip resistance testing
Coefficient of Friction (μ)	0.76 very safe (DIN 51131)
Diesel fuel penetration	Max. 3mm acc. to DAfStb standards

3. Fulfill group of mechanical stress acc. to DIN 18560(1990), Part 7, Table 1

Stress Group	Mobile industrial handling equipment, kind of tyres1)	Stress caused by operational processes and pedestrian traffic
I (heavy)	Steel and polyamide	Processing, grinding and sliding of machinery, placing down of goods by means of metal forks, pedestrian traffic of more than 1000 persons / day

4. Consist of aggregates of varying sizes ranging between 0 to 4mm in diameter, that have been undergone electro-metallurgical smelting processes.

5. Does not contain or pre-bagged with cement binder, that will not harden upon uptake of moisture or water and its shelf time is unlimited.

6. Compared with concrete, the addition of KORODUR 0/4 guarantees and increases:

a. High flexural strength (> 11N / mm²). The higher flexural strength leads to a reduction of the thickness, which is - among others – of special importance for the buildings with fixed load limits and placing thickness

b. Resistant to petrol, mineral oil and solvents

c. Highly wear resistant to vehicular traffic even under the most severe stress

d. Suitable for transporting equipment with steel wheels

e. Suitable for thank traffic

f. Suitable for constantly wet areas

g. Non-skid, anti-slip, even in the case of oil spills and in wet rooms

h. Deicer resistant

i. Non-rusting

j. Non-dusting

k. Physiologically harmless

l. Not electrostatically chargeable

m. Easy to clean

7. Application according to DIN 18353 “Screed works” and to DIN 18560 “Floor screed in building construction”. Besides, cementitious screeds have to be protected from irregular and too rapid drying out.

8. KOROTEX renders, after drying, a glossy protection coat on the surface.
   This means:
   - full hydration
   - prevention of too rapid drying out affecting the quality
   - preventive against contraction cracking tendency
   - closing of pores on the surface
   - improvement of the strength properties in the layer close to the surface

9. Barrier coefficient of KOROTEX is 72% after 3 days and 7 days according to Polymer Institute.

2.03 ANCILLARY MATERIALS

1. Joint Sealants. Refer to relevant specification in this Contract.

2. CONCRETE MIX

   a. Mix and deliver concrete in accordance with the Consulting Engineers Specification.

2.04 FABRICATION

(NOT USED)

2.05 WORKMANSHIP

(NOT USED)

PART 3 EXECUTION

3.01 PREPARATION

A. Concrete substrates must be adequately prepared either by a suitable mechanical method such as scabbling, grit blasting or by such other means as appropriate to remove any cement laitence from the surface. Concrete bases for topping must be carefully prepared to give a clean freshly exposed aggregate surface.

B. Old concrete surface contaminated with oil or grease require suitable preparation such as steam cleaning in conjunction with a suitable detergent. Care must be taken to ensure that the oil or grease is removed from the surface and not simply spread over a larger area. New concrete should be cured for a least 6 weeks according to BS5385 using an approved curing technique e.g. polythene film. Spray-on curing membranes are unsuitable for use on substrates where PCI Emulsion Latex mixes are to be applied subsequently.

C. Roughen existing concrete slab, clean and prepare surface to receive epoxy bonding agent prior to laying screed.

D. Batching

1. Proportions of mixes made with dense aggregates are specified by weight and, where practicable, should be batched by weight. Volume batching will be permitted on the basis of the previously established weight: volume relationship(s) of the particular materials and using accurate gauge boxes. Allow for bulking of damp sand.

2. Proportions of mixes made with lightweight aggregates are specified by volume and should be batched using accurate gauge boxes.

E. Mixing

1. Do not use admixtures containing calcium chloride.

2. Water content of mixes to be the minimum necessary to achieve full compaction, low enough to prevent excessive water being brought to the surface during compaction.

3. Mix materials thoroughly to a uniform consistence. Mixes other than no-fines must be mixed in a suitable forced action mechanical mixer. Do not use a free fall type (drum) mixer.

4. Use while sufficiently plastic for full compaction.

5. Use ready-mixed retarded screed mortar within the working time and site temperatures recommended by the manufacturer. Do not retemper.

F. Adverse Weather

1. Do not lay screeds / toppings unless their surface temperature can be maintained above 5°C for not less than 4 days thereafter.

2. In hot weather reduce the time between operations or use other measures to prevent premature setting or drying out.

G. Joints in screeds: Unless otherwise specified

1. Cast screeds continuously, as far as possible with joints where indicated on drawings and to coincide with joints in finish materials placed over. Form joints with Architect approved joint filler for full screed depth where finish materials are placed over. Where the screed is not to be covered by finish material, provide min. 12mm wide sealant and backing ref Z22.

2. Form day joints with a vertical edge.

H. Levels of Floor Screeds / Toppings: Permissible deviation in level of surface of screeds (allowing for thickness of coverings) and toppings from adtum: +/-10mm.

I. Flatness / Regularity of Floor Screeds: Sudden irregularities are not permitted. When measured with a slip gauge to comply with BS8204 : Part 1, figure 3 or equivalent, the variation in gap under a straightedge (with feet) placed anywhere on the surface to be not more than the following:

1. Screeds to receive toppings or beds 15-30mm thick: 10mm under a 3m straightedge.
2. Screeds to receive mastic asphalt flooring/ underlays: 5mm under a 3m straightedge.
3. Screeds to receive sheet or tile finishes bedded in adhesive:

- 5mm under a 3m straightedge
- 2mm under a 1m straightedge

J. Compaction of Screeds: Compact proprietary screeds using methods recommended by the manufacturer. Compact other screeds as follows:

1. Compact screed layer(s) thoroughly by mechanical means (e.g. plate vibrator) or, where this is not practicable, by hand using a handrammer or weighted roller.

K. Install joint-filler strips where topping / screeding abuts vertical surfaces, such as column pedestals, foundation walls, grade beams, and other locations, as indicated.

1. Extend joint-filler strips full width and depth of joint, terminating flush with topping surface, unless otherwise indicated.
2. Retain subparagraph above or first subparagraph below or, if both are required, indicate location of each on Drawings.
3. Terminate full-width, joint-filler strips 1/2 inch (13 mm) below topping surface where joint sealants, specified in "Joint Sealants," are indicated.
4. Install joint-filler strips in lengths as long as practicable. Where more than one length is required, lace or clip sections together.

3.02 INSTALLATION/APPLICATION

A. SAND CEMENT SCREED

1. Surface must be thoroughly dampened prior to application and excess water removed.
2. PCI Latex Bondcoat to the following proportions:

   (a) Cement : PCI Latex = 1 : 1 by volume

3. Use the minimum clean potable water consistent with workability. Adjust per Manufacturer’s admixture requirement for slurry bondcoat.
4. Mix to a smooth, creamy consistency, adding extra water if required. Apply by stiff brush or broom working it well into the surface.
5. Lay bond coat and top coat in strict accordance with manufacturer’s instructions. Under no circumstances must the slurry be allowed to dry before the new mortar is placed. If large areas of topping are to be laid, the slurry should be applied just in advance of laying. If the PCI Latex Bondcoat does dry it should be mechanically removed and the repair procedure restarted.

6. The topping shall be laid in a single coat application on to the PCI Latex Bondcoat, whilst the bond coat is still tacky.

7. Top Coat : Sand Cement Screed consisting of cement and sand 1:3. Use the minimum amount of water required to achieve workability. The finish to all screed to be a rubbed finish, an even finish formed by a steel trowel.

8. The surface of the screed shall be finished, level, with a steel trowel to give a smooth untextured surface. Maximum deviation permitted in surfaces will be 3mm from a 1800 mm straight edge. All angles to vertical surfaces to be right angles unless stated otherwise.

9. Lay bays in 15 sqm maximum, with a bay length of not more than 1.5 times the width in a chequer board pattern. Allow 24 hours minimum between laying adjoining bays.

10. Immediately after laying, protect surface of floor screeds and in-situ finishes from wind and sunlight.

11. Cure the screed surface by covering with hessian sacks and thoroughly soaking with water. These sacks shall be kept wet with water for the first 3 days after screed placement. Alternatively, cover the screed surface with polythene sheet, tied down at all edges.

B. **WATERPROOF SAND CEMENT SCREED**:

1. Concrete surfaces must be thoroughly dampened prior to application of BASF Febond SBR Slurry Bondcoat, and excess water removed.

2. Mix Febond SBR slurry Bondcoat of 1 part Febond SBR to 1 part of cement by volume. Mix to a smooth, creamy paste.

3. Brush apply two full coats of this slurry, to a total thickness of 1mm approximately, allowing the preceding coat to dry (usually overnight) prior to applying the second at right angles to the first.

4. When the slurry coats have dried, apply a third coat (at right angles to the second) and immediately render or screed onto it. Apply min. 13mm screed to horizontal surfaces, or two coats of render to vertical surfaces (to min. 13mm total thickness), to tank the area completely.

5. When the screed has dried sufficiently to take foot traffic, apply fillets of the same mortar at floor/wall junctions and internal wall angles.

6. Mortar mixes for screeds, renders and fillets are to be 3:1 sand cement, + 10 litres Febond SBR per 50kg cement. Add water as required for workability.

7. For high water pressure, increase Febond SBR use to 15 litres per 50kg cement. (In this case sand must not be too wet or desired workability may be unattainable.)

8. All day joints in slurry and/or render should be staggered and slurry coats must be lapped.

C. **Dry shake Floor Hardener**

Straightly follow the instructions from the manufacturer.

D. **Roof Membrane & Insulation (where applicable):**

1. Refer to Section 6.1 and 6.2

E. **Upper Concrete Topping**:

1. Ensure that roof insulation boards are closely lapped at all rebated edges so that gaps are not created during the laying of the upper protection topping. The Contractor and the Roof Manufacturer's Technical Representative shall inspect

all roofs and decks to ensure that the insulation board installation is without defects and is ready for the application of the upper topping layer. Commencement of upper topping installation shall constitute acceptance of insulation board substrate.

2. Follow mix and handling procedure for cement/sand screed mixed with PCI Emulsion as previously described above, except as noted in this Clause. Place upper topping in two layers. Lay the bottom topping layer over the rigid insulation board, compact bottom layer. Rake its surface to provide a bond for the second layer. Install steel wire fabric reinforcement over the entire bottom screed layer install top layer in a continuous operation. Total thickness shall be 85mm. (Discontinue fabric reinforcement at all expansion joint locations.)

3. Install a 15 mm thick topping layer mixed with granite fines in a 1:4 ratio laid monolithically. The topping layer shall be mixed also with PCI Emulsion to provide water-resistance and hardness to layer.

4. Form and install expansion joints at the following locations :

   a. Along all perimeter edges against parapets and curbs, around abutments and roof penetrations.

   b. Along transitions between different roofing materials or substrates.

   c. At all expansion, control, construction, cold and seismic joints in the roof deck structure.

   d. At 3.6 metres on center at each direction in the field of all roofs and decks. In addition, form and install control joints in roof screed at 1.2 metres on centres at each direction.

   e. At locations indicated on the approved roof shop drawings.

   f. Install compatible joint filler at full depth of Roof System Assembly as indicated on the approved roof shop drawings. Install bond breaker and sealant.

5. Ensure that upper topping is thoroughly compacted throughout its depth, especially at corners and edge. Allow screed to harden sufficiently to prevent damage when laying subsequent bays. Lay screeds in bays of 15m (maximum) with length and width of 3.6 metres maximum in chequered board patterns. Allow 24 hours minimum between laying adjoining bays.

6. Protect topping and screed during curing as required in Clause 3.04 B-4 above.

F. **Placing Concrete:**

1. Place concrete in accordance with the Consulting Engineers specification.

2. Ensure reinforcement, embedded parts, etc. are not disturbed during concrete placement.

3. Do not deposit concrete which has partially set, hardened or contaminated with detrimental materials.

4. Excessive honey comb or embedded debris in concrete is not acceptable.

5. Do not remove forms until concrete has attained sufficient strength. Formwork clamps or ties may be loosened 24 hours after concrete is placed.

G. **Curing Interior Surfaces:**

1. Cure floor surfaces in accordance with ACI 308.

2. Chemical Cure for Slabs (ACI 308): Apply in accordance with manufacturer’s single coat application instructions. Obtain from the curing compound manufacturer a written warranty that the compound will not be detrimental to

bonding of flooring adhesives or surface materials. Submit warranty to the Employer at the time request is made for use of curing compound.

3. Moisture Cure: Moisture curing method is optional for all areas receiving special flooring (if specified) or hardener.

   a. Place wet, moisture-containing fabric covering as soon as concrete had hardened sufficiently to prevent surface damage.

   b. Cover entire surface, including edges and sides.

   c. Keep coverings continually moist so that a film of water remains on concrete surface throughout curing period. Maintain concrete in moist condition for not less than seven days after placement.

3.03 CLEANING

A. At completion of the Cementitous Floor and Roof Screed Work, remove all temporary protection and barricades from all decks and roofs, and the Project Site.

3.04 PROTECTION

A. Clean all items and adjacent surfaces of work. Examine all floor and roof screed damage. In the event damage is irreparable, remove and replace such items at no additional cost. Leave entire Work in condition acceptable to receive specified finishes.

B. The Contractor shall at all times keep the Project roof and slab area free of excessive waste, materials and rubbish. Prior to completion of all Cementitious Floor and Roof Screed Work, the Contractor shall remove all such waste, equipment, other property and tools from the Project Site. Legally and safely handle, transport, and dispose of waste and debris.

C. The Contractor shall continually protect the Cementitious Floor and Roof Screed System from all damage and traffic during and after installation work until Project Handover. Install and provide temporary traffic pads on screed areas to protect from work of other trades during Project Construction.

END OF SECTION